

The Fundamentals of NAND Flash Memory



Technology for tomorrow's fourth industrial revolution

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With the rise of the mobile-centric era, data-driven applications such as the Internet of Things, artificial intelligence, cloud computing, blockchain, and so on are ever increasing. As these applications create unprecedented amounts of data, the technologies used for collecting, processing, and storing them well are becoming more crucial. According to the research in [1], the amount of data we generate has a growth rate of 2.5 times per every four years, and the total amount of data will reach 175 ZB by 2025. Such a “Big Bang” of data means a tremendous

demand on storage systems; it is evident that high-capacity and high-performance storage systems will play a key role. Therefore, along with this trend, NAND flash memory technology has continued to evolve to meet high demands. This article covers key technologies for high-density/high-performance NAND flash memory.

This article will cover key technologies for high-density/high-performance NAND flash memory. The section, “What Is Flash memory, and How Does It Operate?” explains basics such as operation principles, and the “How to Best Implement an NAND Flash Cell Array” section describes related key circuits and processes. Then, the “History of the NAND Flash: A Bird’s-Eye View” section covers a brief history of NAND flash

development, from its beginning to state-of-the-art technologies. Moving on, the “Future Trends” section will explore candidates and trends for future NAND flash.

What Is Flash Memory, and How Does It Operate?

Figure 1(a) illustrates the structure of a flash memory cell. A typical MOS transistor consists of gate/source/drain and has a fixed threshold voltage. On the other hand, when a storage node (e.g., floating gate) has been newly added to a flash memory cell, its threshold voltage is controlled by the number of electrons injected into or ejected from the node. Once a state (the total number of electrons) is determined it cannot be changed, even without a power supply as this

Digital Object Identifier 10.1109/MSSC.2022.3166466
Date of current version: 25 June 2022

node is electrically isolated, which gives the cell a nonvolatile feature.

One of the most well-known principles for injecting or ejecting electrons into the storage node is Fowler–Nordheim (FN) tunneling [2]. As shown in Figure 1(b), tunneling occurs in an oxide placed between a substrate and a storage node. If a sufficiently high E-field is applied to the oxide, the rectangular-shaped barrier gets deformed to a triangular shape. As a result, the distance electrons must travel for tunneling becomes shorter, and finally, the tunneling current is generated. The operation of injecting electrons into the storage node is called a *program*. To program a cell, the direction of the E-field should be from top down so that high voltage (V_{Program}) is applied to a gate, and ground voltage is applied to the source/drain and the substrate. When the program operation is done, more higher $V_{\text{Control_Gate}}$ voltage is required to turn on the transistor to overcome the electrons trapped in the storage node (i.e., the threshold voltage V_{th} of the cell increases), and its gate voltage (VG)-drain current (I_d) curve moves from left to right [Figure 1(c)]. The opposite action, which ejects elections from the node, is called an *erase*, which effectively “lowers” the V_{th} of the cell.

Unlike dynamic random-access memory (RAM) or static RAM, the flash memory cell has a limited life span [3]. Other nonvolatile memories, such as resistive RAM (ReRAM) [45] and phase-change RAM (PCRAM) [46], also have this characteristic. The reason is as follows. If FN tunneling for program and erase (P/E) is repeated (the P/E cycle), side effects start showing up, where unwanted electrons are trapped inside the oxide or at the interface [see Figure 2(a)]. When this happens, the V_{th} of the cell is changed to a direction we do not want (generally, it increases). As the P/E cycle is repeated, the window between two states is gradually narrowed down and eventually reaches a dead point. This property is known as *endurance*. Trapped charges also cause other issues, such as retention. As shown in Figure 2(b), charges are

thrown away as time goes by, and this also leads to a degradation of operation margin or device failure. This property is referred to as *retention*, and these two are known as the *reliability characteristics of flash memory*. As having competitive technologies for overcoming reliability issues are one of the important factors in determining the completeness of a flash memory product, many ideas have been proposed in various fields, including materials, process integration, circuits, and system-level management [4]–[42].

How to Best Implement an NAND Flash Cell Array

In this section, the unique structural points that only an NAND flash cell

array has are observed. As presented in Figure 3(a), cells (green color) are connected in series, and the selectors, called the *string select line (SSL)* and the *bitline (BL)*, are connected at one end through the BL contact, and at the other end, called the *ground select line (GSL)* and the *common source line (CSL)*, are connected. This structure is referred to as a *string* (pink color). If one looks at this structure closely, it is clear that it resembles a logic gate NAND’s pull-down network, and that is the reason it is named an *NAND flash*. Several strings are placed in parallel along an X-direction, and gates of the cells at the same position in a string are connected to each other by one common

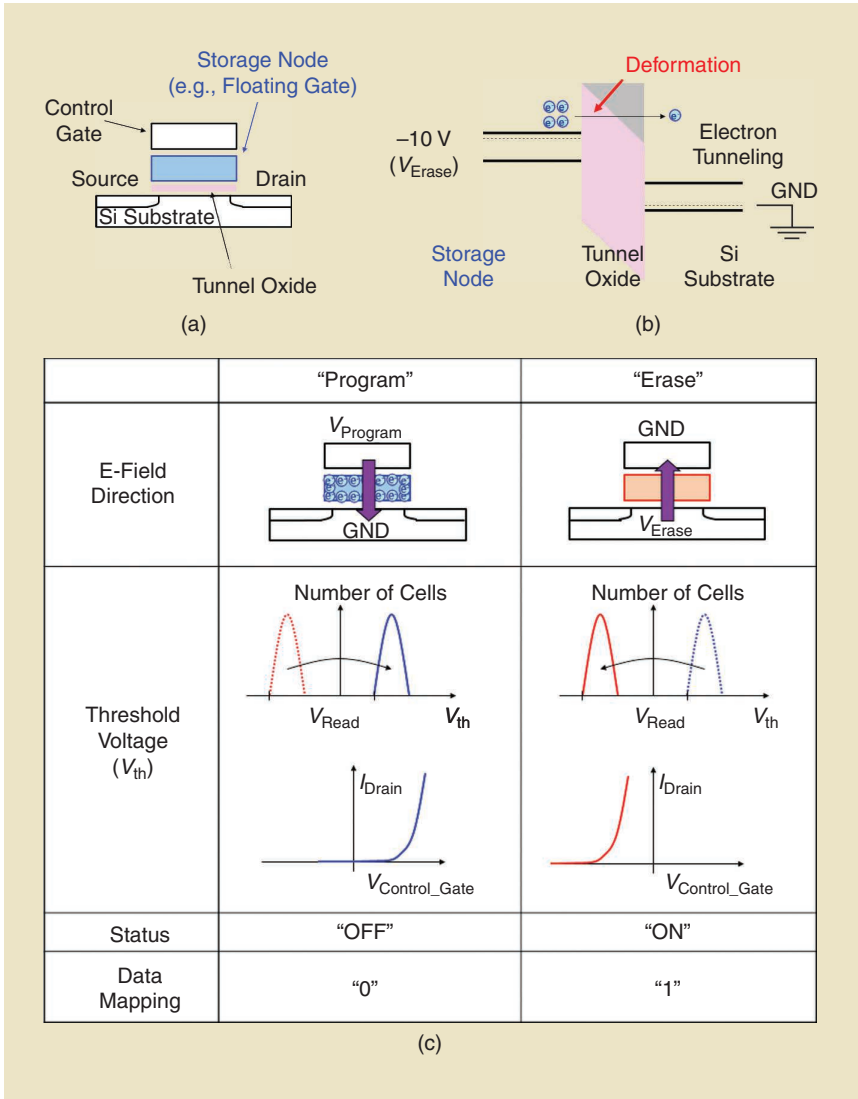


FIGURE 1: (a) The structure of a flash memory cell, (b) Fowler–Nordheim (FN) tunneling, and (c) a program-and-erase operation. Si: silicon; GND: ground.

word line (WL). This structure is known as a *page* (colored in blue), and finally, a set of strings is called a *block* (brown box).

If one looks at the cross-sectional view of a string [Figure 3(b)], it will be clear that there are only two contacts, one each at both ends of the BL

and CSL [10]. In addition, no contacts are used to form a page; they are connected using a WL. Finally, all the cells in a block share a single-pocket p-well. Thanks to these unique structural advantages, it is possible to realize a cell size close to $4F^2$, which is the ideal lower limit of a memory cell size; thus, an NAND flash memory shows superior areal density and incomparable cost competitiveness when compared to other types of memory.

History of the NAND Flash: A Bird's-Eye View

In the previous section, the basics of an NAND flash memory were briefly covered. Here, an examination of the history of NAND flash technology, from its invention, to the present, and some of its defining moments, is now offered.

Figure 4 depicts the 38-year-long history of NAND flash memory. The data are extracted from papers published in the IEEE International Electron Devices Meeting, International Solid-State Circuits Conference, VLSI Circuits Symposium, and several other meaningful events [4]–[42]. The whole period is divided into three major parts: the first is its invention; the second, 2D NAND; and the third, 3D NAND. The basic operation concept of flash memory was presented for the first time in 1984 [4]. In 1995, NAND flash finally succeeded in commercialization [8]. The first product was a 32-Mb, single-level cell device based on a 470-nm process. Six years later, 2D (planar) NAND surpassed 1 Gb in its density [11], and multilevel cell (MLC) products that store 2-bit per cell were successfully deployed [10]. After a rapid evolvement of more than 10 years, 3D vertical NAND (V-NAND), a disruptive technology, first appeared on the market in 2014. This history of continuous innovations was possible with the help of numerous breakthrough technologies in various areas such as materials, process integration, circuits/algorithms, and system-level management.

First Phase: Invention

In 1984, the concept of “Flash EEPROM” was first presented by Masuoka et al.

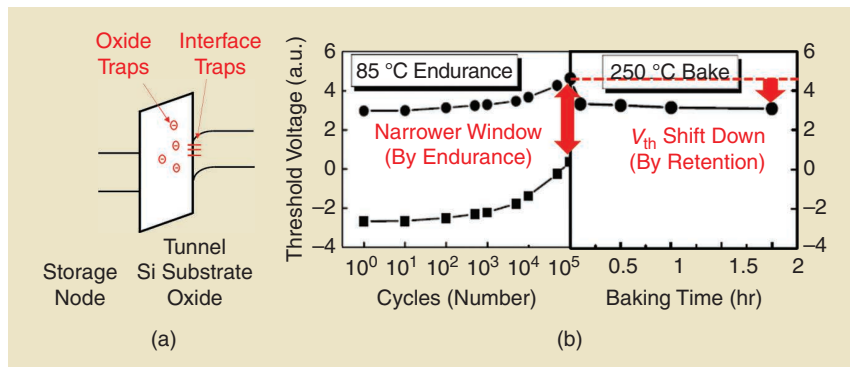


FIGURE 2: The reliability issues of a flash memory cell. (a) The trap/detrapp of unwanted electrons and (b) operation window degradation by reliability issues. Si: silicon.

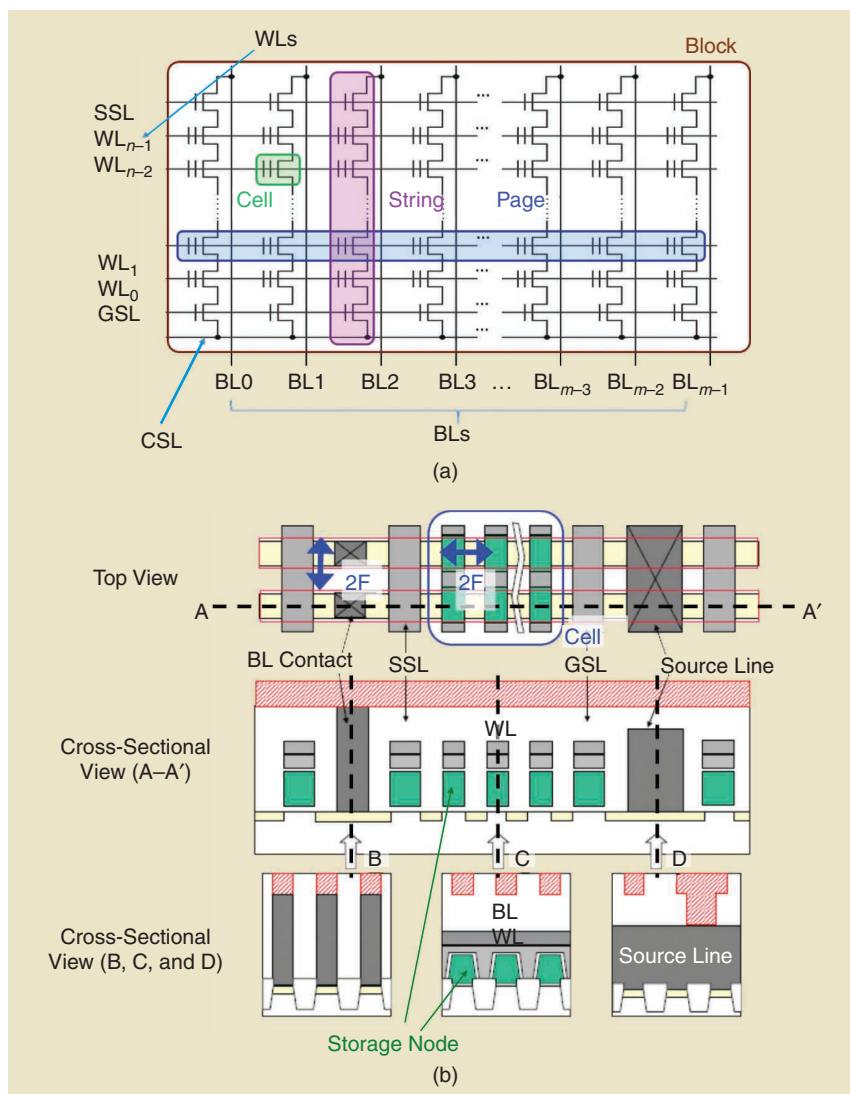


FIGURE 3: (a) The string, page, and block of an NAND cell array and (b) a cross-sectional view. WLs: word lines; CSL: common source line; BLs: bit lines; SSL: string select line; GSL: ground select line.

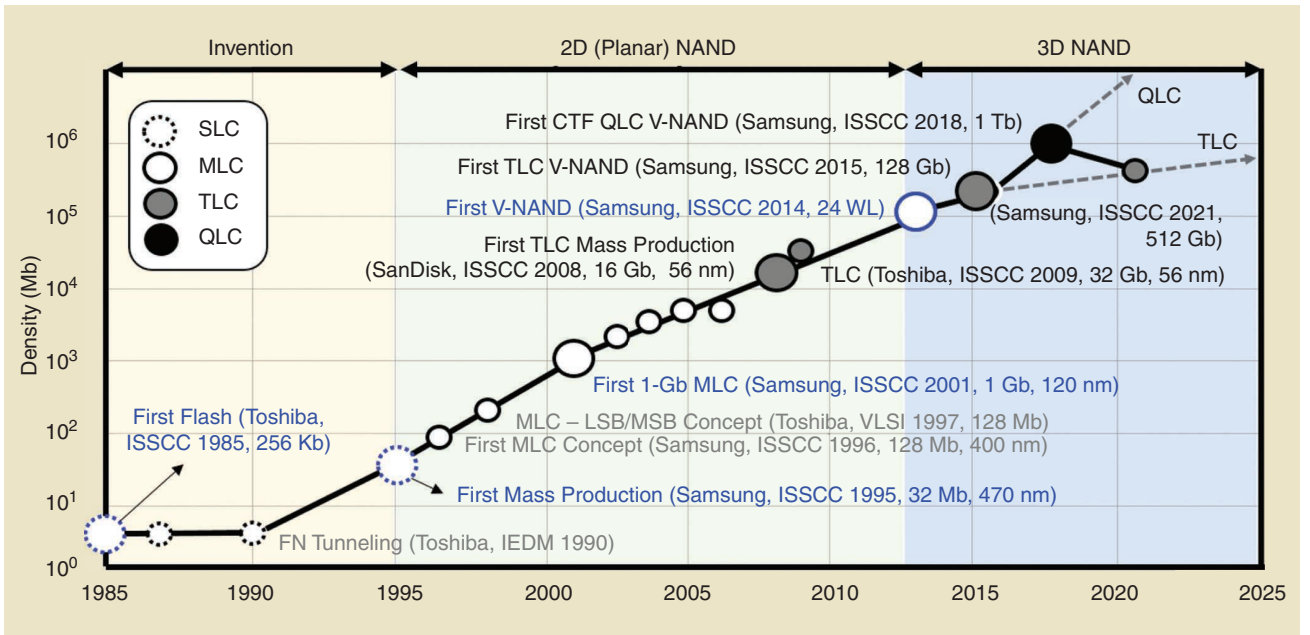


FIGURE 4: The history of NAND flash development: a bird's-eye view. SLC: single-level cell; QLC: quadruple-level cell; CTF: charge-trapped flash; MLC: multilevel cell; TLC: triple-level cell; V-NAND: vertical NAND; ISSCC: International Solid-State Circuits Conference; VLSI: VLSI Circuits Symposium; IEDM: IEEE International Electron Devices Meeting.

[4]. The proposed EEPROM cell consists of three terminals: a floating gate for data holding, control gate for cell selection, and a gate for erase operations. We should note that erase gates of cells are connected in common. As a result, all of these are erased together at the same time, i.e., all data are gone in a “flash” moment. This is the reason Masuoka named this device *flash*. He also demonstrated a chip with 256 Kb of density based on a 1.2- μm design rule in 1985 [5]. A few years later, the concept of an NAND string structure was presented for the first time [6], [7]. It featured two select transistors for a string, and a double-well structure for noise separation between a cell array and a peripheral circuit.

Although many technological achievements had been made since its first appearance, there were two crucial hurdles to overcome for commercialization. The first one was a high-voltage (HV)-related issue: nearly ~ 30 V should be directly applied to WLs and BLs. Therefore, flash cells and related selectors were not able to be scaled down. In addition, the circuit size needed to generate the corresponding voltages had to be large, which played as a major hurdle in increasing the

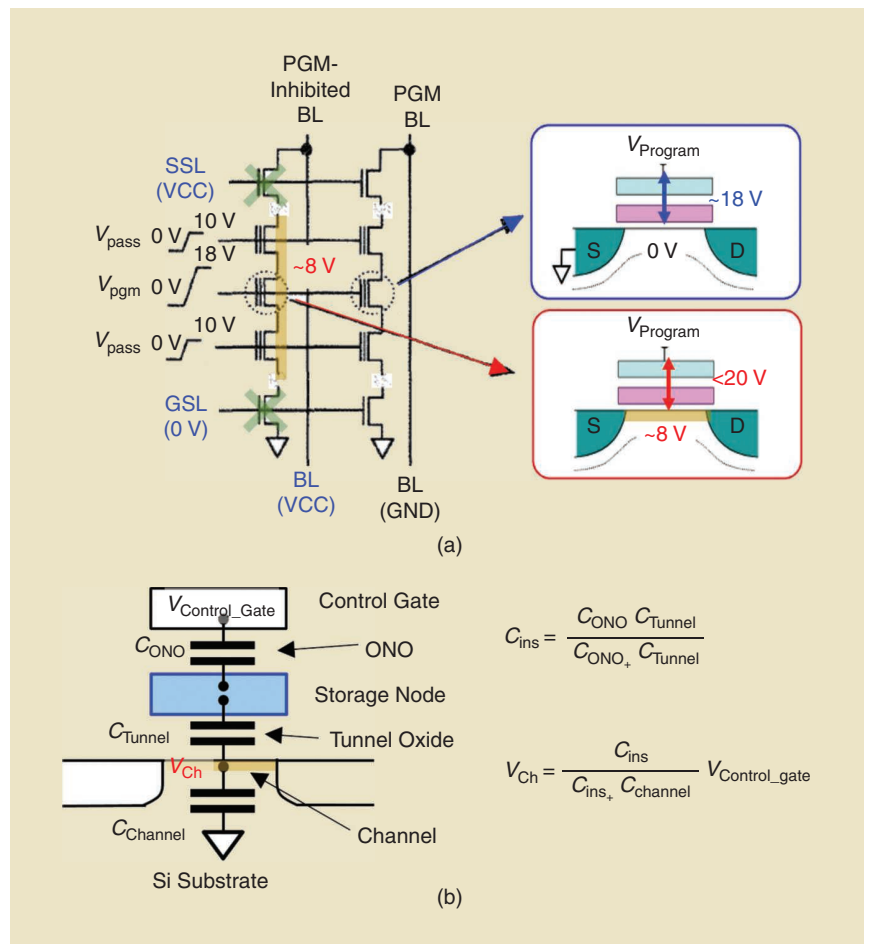


FIGURE 5: A self-boasting scheme. (a) A newly defined operation bias and (b) the concept of capacitive coupling at a channel. PGM: program; GND: ground Si: silicon; S: source; D: drain.

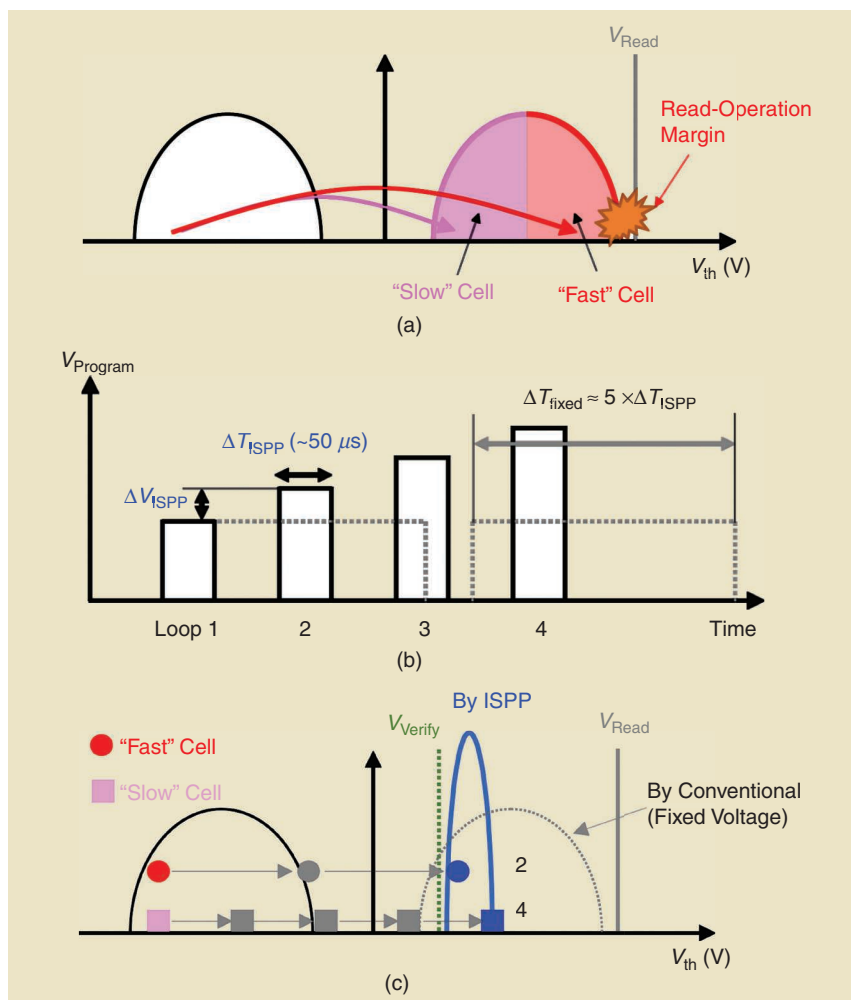


FIGURE 6: An ISPP scheme. (a) The threshold voltage distribution after a program operation, (b) a series of gradually increasing program pulses, and (c) the behavior of "fast" and "slow" cells.

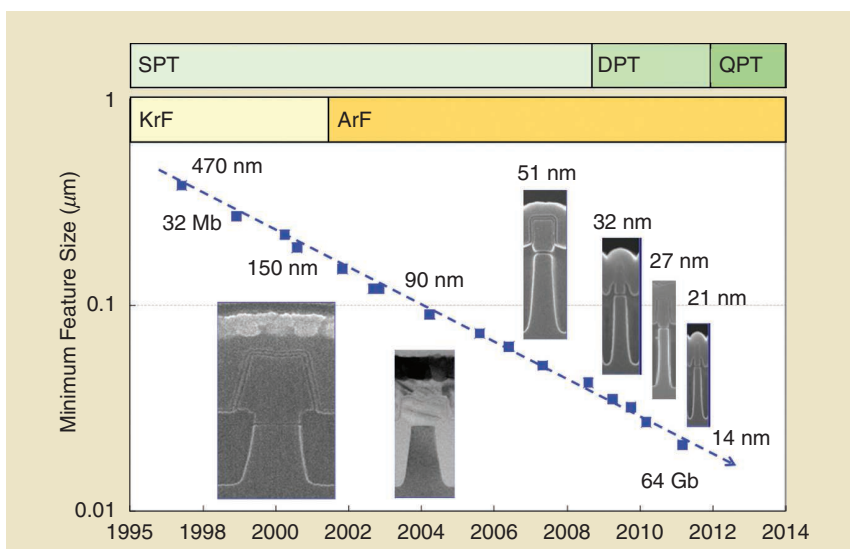


FIGURE 7: The scaling trend of a 2D NAND flash. SPT: single-patterning technology; DPT: double-patterning technology; QPT: quadruple-patterning technology; KrF: krypton fluoride; ArF: argon fluoride.

bit/areal density and reducing the cost of the device. Second, it was so hard to make the width of the V_{th} distribution tight while satisfying the writing-performance requirement; a narrow distribution is essential for NAND flash to secure a minimum operation margin, but at that time, it was only achieved by an unacceptable degree of sacrifice in its performance.

These critical issues were tackled by two key techniques [8]. The first one was a self-boasting scheme. As illustrated in Figure 5(a), only $\sim 0 V_{CC}$ is used to control BLs, SSLs, and GSLs, and the required high voltages are not given from the external HVdc supply but internally generated by so-called "self-boasting," a capacitive coupling operation between a control gate and a channel [Figure 5(b)]. Thanks to the idea, the following became possible: 1) implementation of a small chip due to its smaller page-buffer (PB) size for BL control, 2) performance enhancement by utilizing BL-operation parallelism (by integrating more small-size PBs in the chip), and 3) scaling down of the cell array (cell/SSL/GSL), which, in turn, lowered the cost of implementation by reducing necessary process steps.

The second one was an incremental step-pulse programming (ISPP). The program speed of each cell is inherently different, even under the same write condition (temperature and program voltage); slow or fast due to nonuniformity in their physical dimensions such as thickness of the tunnel oxide. Therefore, the overall write performance is limited by the slowest cell. A higher performance cannot simply be achieved by applying higher, fixed-level program voltage for slow cells because this can result in an overprogramming problem; the read operation-margin issue is due to faster cells [see Figure 6(a)]. In an ISPP scheme, a series of program pulses with a relatively short duration ($\sim 50 \mu s$, approximately one-fifth of the conventional pulse) and gradually increasing levels is applied sequentially [see Figure 6(b)]. The proposed scheme effectively covered the speed variation by allowing fast cells to be programmed with a

lower program voltage, and slow cells to be programmed with a higher program voltage, which resulted in much narrower V_{th} distribution. As depicted in Figure 6(c), the fast cell was sufficiently programmed at loop 2. After that, it is automatically switched to a “program-inhibit” state so that the cell is not affected by subsequent higher voltages for the slow cell programming anymore. As a 1-V increment of the voltage was proven to be approximately as effective as a five-times increment of the time duration with a fixed voltage level [8], [9], the time needed for completing the programming operation became shorter, which resulted in a high writing performance.

Second Phase: 2D (Planar) NAND, a Period of “Big Bang”

With the help of these two key enablers, 2D (planar) NAND entered a period of full-scale progress—known as a period of *Big Bang*. During this time, the density of an NAND device was doubled every two years (see Figure 4). These advances had been driven by two major technology trends: continuous cell scaling and multibit/cell technology.

Figure 7 shows a trend of cell scaling. Cell size had been reduced to one-thirtieth, and the device's density had increased by more than 1,000 times over 20 years. The first product was based on 470-nm technology and had 32-Mb density [8], but around 2014, 128-Gb devices based on 16 nm were commercialized [22]. Advances in lithography were one of the main drivers for this trend. ArF-based lithography processes started to be adopted from ~100-nm technology nodes. In addition, double-patterning technology was deployed from ~30-nm node to overcome the technical limitation of single patterning, while maintaining the production cost at a reasonable level [20].

Along with cell scaling, multibit technology also played a key role. MLC technology, which stores 2-bit in one cell, became mainstream in the marketplace in the early 2000s [12], and triple-level cell (TLC), 3-bit-per-cell technology was launched circa 2008 [17]. As TLCs could achieve up to a 50% more areal density

compared to MLCs in theory, they were accelerators for more bit/areal density and cost/bit reduction.

The second trend was driven by many innovative ideas from various fields for overcoming the cell-to-cell interference phenomenon, which is definitely one of the most serious

problems in NAND flash. As displayed in Figure 8(a), the unwanted electrical influence for each other becomes worse as the distance between adjacent cells becomes shorter by scale down. Number of electrons stored in a cell is also dramatically reduced. For example, the amount of interference in

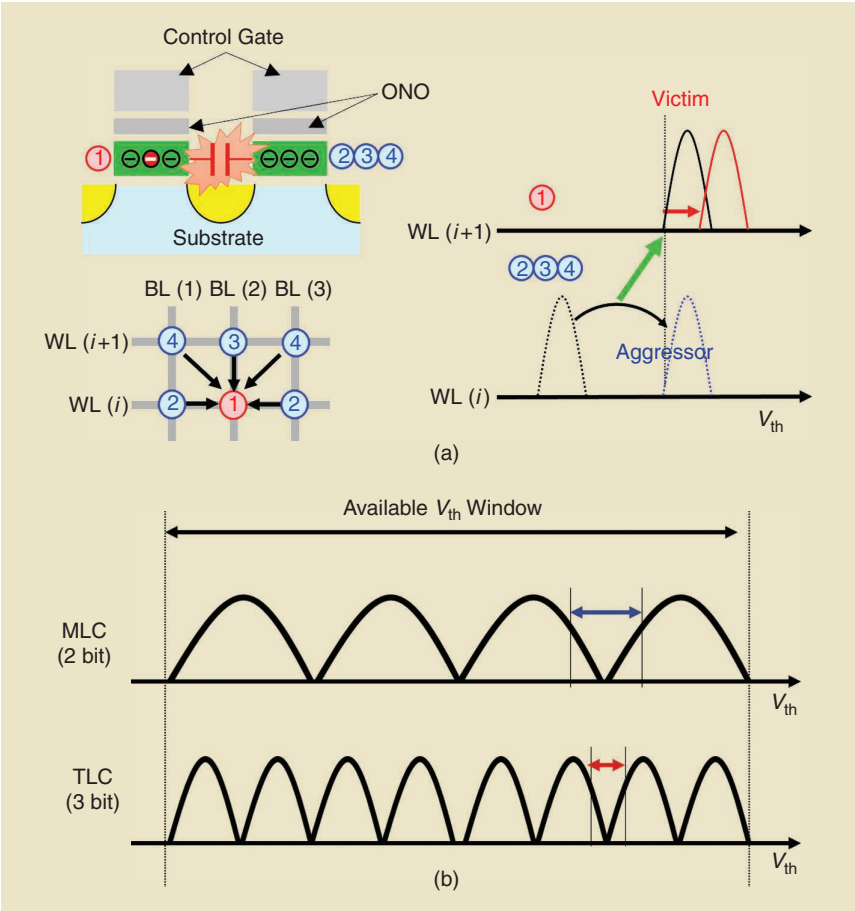


FIGURE 8: (a) Cell-to-cell interference. (b) A comparison of operations for MLC and TLC.

TABLE 1. TECHNIQUES FOR OVERCOMING INTERFERENCE.		
TECHNIQUES	INTERFERENCE DIRECTION	FIRST SHOWN (AT PRODUCT)
Shadow program [10]	WL	63-nm MLC
All-BL program [16]	BL	—
Dual-pulse program [23]	BL	—
Data randomizing [25]	WL/BL/diagonal	35-nm MLC
Data recovery [15]	WL	—
Negative V_{th} control [19]	WL/BL/diagonal	27-nm MLC
W-gate, BL air gap [20], [21]	WL/diagonal	—
Reprogram [14]	WL/diagonal	27-nm TLC
OBP [24], [26]	WL/BL/diagonal	—
OBP: on-chip buffering.		

43-nm node was increased by 2.4 times, and the capacity for storing electrons was reduced down to one-tenth, or only a few hundred of electrons, compared to those of a 160-nm device [19]. The situation becomes worse when it comes to a multibit device:

the operation margin for a TLC was diminished to just half of an MLC as four more states should be formed in within the same available V_{th} window [Figure 8(b)].

Some of the key ideas for overcoming interference are summarized

in Table 1. At the device level, there have been innovations in materials, sophisticated processing, and patterning, such as W-gate and BL air gap [20], [21]. At the circuit and system level, many programming algorithms, biasing methods, and system-level data management techniques have been proposed [10]–[26]. Shadow program techniques [10], in which the programming sequence have been carefully redesigned to mitigate interference in the WL's direction, played a true enabler in opening the MLC era, and became the de facto standard write algorithm for 2D NAND. The techniques described in [16] and [23] also contributed to suppressing interference from the BL direction. Along with efforts at the discrete NAND-chip level, a brilliant idea was proposed from the storage-system side [25]. In this method, an NAND controller in the system encodes an original data set before writing it into the NAND. With the help of this operation, a certain degree of randomness is added to the data set. As a result, the probability for occurrence of the worst pattern in terms of interference could be dramatically reduced. Finally, the operation margin issue in TLC was overcome thanks to ideas such as on-chip buffering [24], [26] and reprogramming [14] while keeping the performance gap with MLCs as small as possible (typically <2.5 times).

Third Phase: 3D (Vertical) NAND, the Era of Revolution

Even with the numerous efforts described in the previous section, 2D NAND technology eventually reached a dead end due to 1) skyrocketing production costs as a result of extremely complex processes, such as quadruple patterning becoming mandatory in an ~10-nm-era and 2) it no longer being possible to meet performance requirements because of more seriously deteriorating reliability and interference characteristics. For these reasons, there was an urgent need for a “revolution.”

At that time, 3D NAND had emerged as a disruptive technology. In fact, 3D NAND has been researched since 2006 [27]. It simply stacked two 2D NANDs

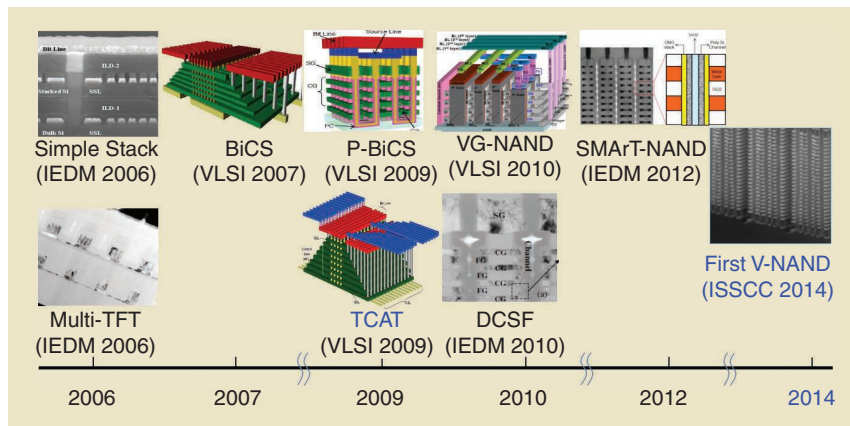


FIGURE 9: Various 3D NAND types. ISSCC: International Solid-State Circuits Conference; VLSI: VLSI Circuits Symposium; IEDM: IEEE International Electron Devices Meeting; BiCS: bit-cost scalable; P-BiCS: Pipe-BiCS; SMARt-NAND: stacked memory array-NAND; DCSF: dual control-gate with surrounding floating-gate; TFT: thin film transistor.

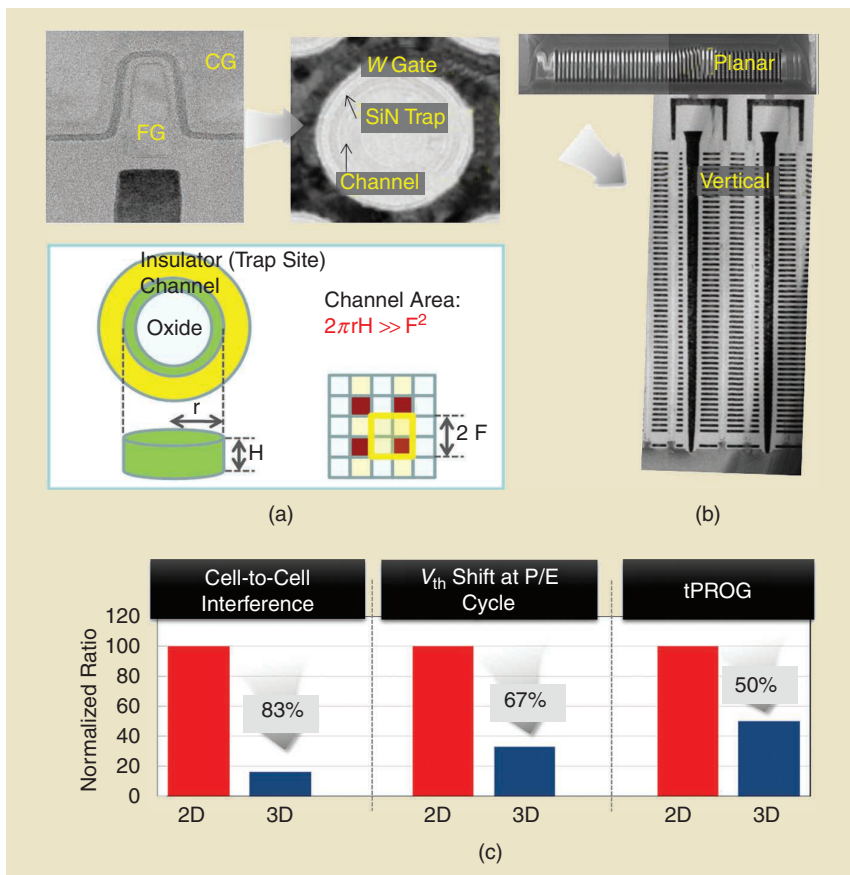


FIGURE 10: A charge-trapped flash (CTF)-based V-NAND cell. (a) A cylinder-type structure, (b) a vertically stacked string, and (c) superior cell characteristics. SiN: silicon nitride; CG: control gate; FG: floating gate; W gate: Tungsten gate; tPROG: program time.

and connected them using through-silicon via. It is easy to imagine that it's not a true 3D V-NAND, so it could be called 2.5D. Full-scale ideas started to be proposed in 2007. Going through various ideas such as bit-cost scalable [28] and VG-NAND [30], terabit cell-array transistor (TCAT) technology was commercialized for the first time in 2014 [29], [32], [34] (Figure 9).

As displayed in Figure 10(a), a TCAT cell is a charge-trapped flash (CTF) that uses a silicon nitride trap as a storage node, which is different from that of a planar-floating gate [29]. The channels are formed in a hole shape and surrounded by a tungsten (W) WL. In this structure, the interference between adjacent BLs becomes zero because the WL is completely blocked. Interference in the WL direction is also dramatically alleviated compared to conventional, floating gate cells thanks to its CTF structure, which is better in terms of interference in nature. When it comes to reliability, a TCAT is superior because the effective area of a channel is much larger than that of the planar counterpart, and the operating voltage for the P/E is lowered by tunnel-barrier engineering [34]. Finally, faster read/write operation is possible with the help of its low resistance of metal (W) WL and excellent reliability characteristics. Figure 10(b) compares string structures. As shown, the string placed on a 2D plane is erected in a Z-direction by vertically stacking WLs. Therefore, the technical challenges in a 3D NAND come from forming 3D structures such as channel-hole drilling and WL etching, not from sophisticated lithography for 2D patterning. By fully utilizing the many structural advantages described previously, the 3D NAND shows a much better performance [33]: interference was reduced by 83%, a V_{th} shift after the P/E cycle by 67%, and a write performance by more than 50% [see Figure 10(c)]. As a result, it has been nearly 10 years since 3D NAND TLC devices, with their superior bit areal density, cost, and performance, replaced MLC 2D NAND. Moreover, even quadruple-level cell (QLC) technology has been successfully deployed to market [38]. Regarding QLC

implementation, the following should be considered: 1) The performance gap should be narrowed down. As depicted in Figure 11(a), the write throughput of a QLC is only one-quarter that of a TLC, and the gap is becoming smaller. 2) When it comes to operation margin, the trend is similar [see Figure 11(b)]. Therefore, more advanced design techniques are strongly required, and much research is still ongoing today [38], [40], [41].

As the generations evolve, 3D NAND is also facing a new type of

technological hurdle. A product with up to 176 stacks was introduced to the market [43], and its upward trend is expected to be roughly +30% in its number-of-stacks per generation. In addition, a unit-mold pitch (a height of one-WL stack) is expected to be scaled down for approximately 10% per generation. As 3D NAND stacks up more WLs, critical limitations are found in channel-hole etchings, mold leanings, and warpages. Meanwhile, performance/reliability degradation due to interference is becoming relevant again

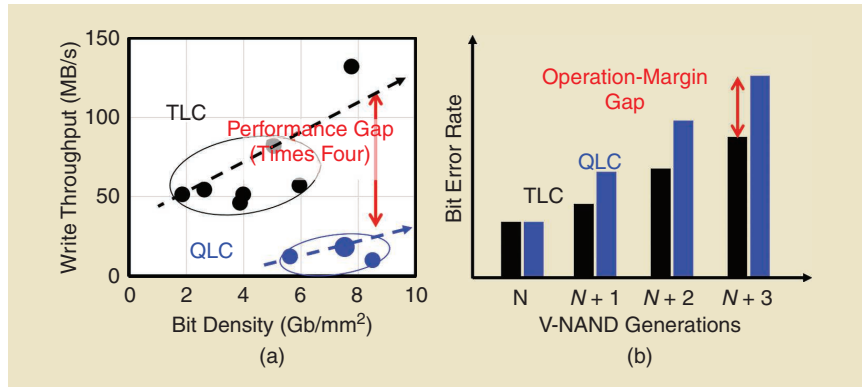


FIGURE 11: The technical challenges in QLC implementation. (a) The performance and (b) operation margins.

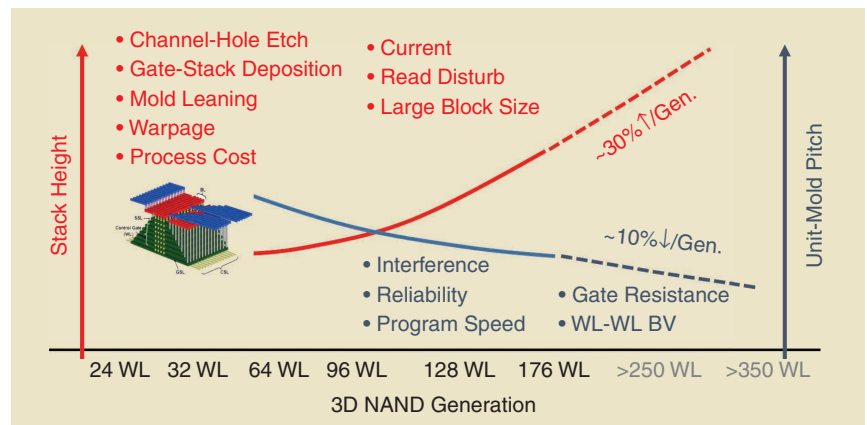


FIGURE 12: The technical hurdles for future 3D NAND. Gen: generation; BV: breakdown voltage.

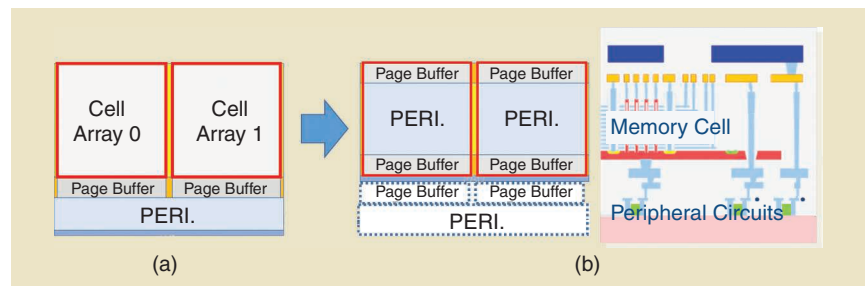


FIGURE 13: A COP structure. (a) A conventional non-COP and two-plane architecture and (b) a COP with a four-plane architecture. Peri.: peripheral.

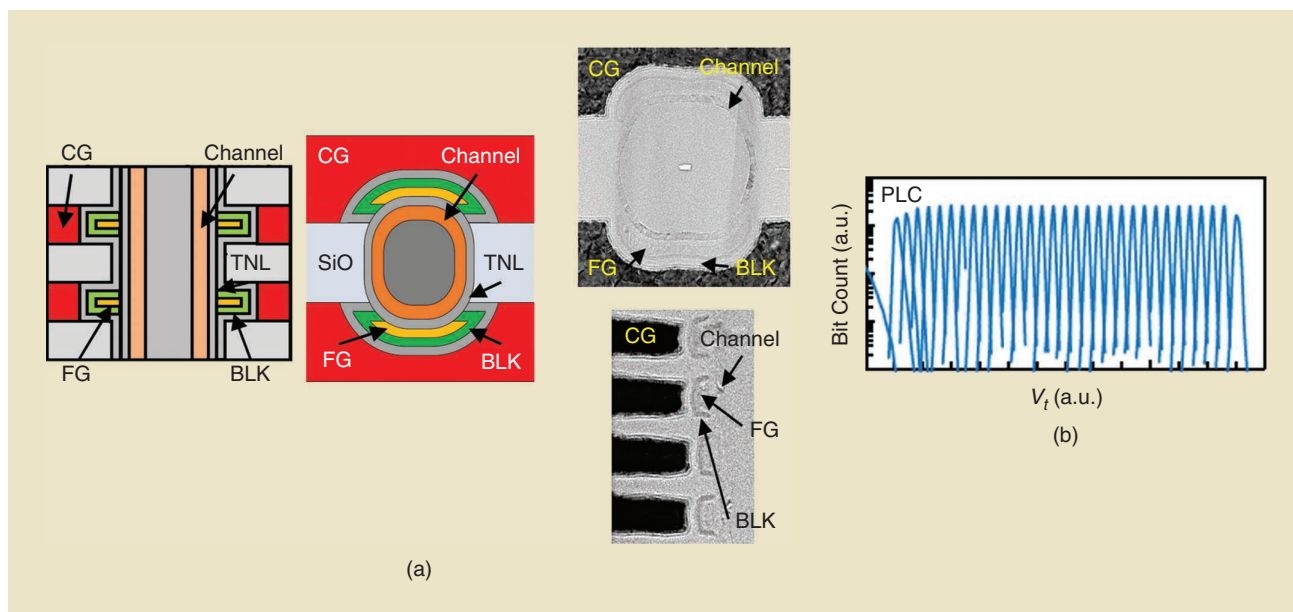


FIGURE 14: The future 3D NAND candidates for higher areal density. (a) A split-gate cell and (b) programmable logic controller (5-bit per cell). Si: silicon; SiO: silicon dioxide.

as mold height is decreasing. Continuously increasing manufacturing costs is also a big issue. The challenges 3D NAND is facing are briefly summarized in Figure 12. Further discussions on this subject can be found in [44].

Future Trends

From now on, let us take a look at future candidates of NAND flash. State-of-the-art 3D NANDs are increasingly adopting cell-on-peri (COP) technology [41], [42]. As shown in Figure 13, it is literally a technology that places a cell array “on” a peripheral circuit and connects it using through-hole vias. In this case, the chip size is dramatically reduced as most of the peripheral circuit, whose area portion is roughly 20~30%, can be hidden. In addition, enhancing performance becomes easier because more circuits can be added under the array. For example, most COP-based 3D NAND devices are adopting four-plane architectures by adding more PBs at the vacant area under the cell array. In this case, the read/write performance of the COP-based device is doubled, compared to conventional two-plane devices. However, the problem is the manufacturing cost. As the peripheral circuit and cell array must be implemented separately, the total number of process

steps should be increased by at least 30%. The increased design complexity also poses hurdles to be overcome. The place-and-routing plan should be carefully designed to efficiently connect the upper and the lower circuits while satisfying several constraints such as signal integrity (time delay, voltage drop, and crosstalk). Finally, new issues such as mechanical stress must be addressed adequately.

Along with COP, technologies for increasing bit area density are being continuously researched. Split-cell technology, introduced in 2019, is a good example [43]. It physically splits the existing cylindrical cell in half and searches for the possibility of a 5 bit/cell, as illustrated in Figure 14. If this technology succeeds, a drastic jump in terms of areal density can be achieved; however, many new technical challenges are anticipated. For example, an unprecedented level of error-correction capability may be mandatory for a NAND controller to overcome a near-zero-operation margin among V_{th} distributions, which may result in side effects on the total power and cost of a storage system.

Conclusion

In this article, the 38-year journey of NAND flash, from its beginning to its possible future, was presented by

touching on some defining technical challenges and innovations the NAND industry has undergone thus far. It is obvious that NAND flash, as it has done previously, will play a key enabler in the upcoming data-centric era. This means that the future prospects of the NAND industry is very positive, and its upward trend is expected to accelerate. Therefore, I can confidently proclaim that now is the time to prepare another disruption, just like 3D NAND did eight years ago. The possibilities are truly endless and widely open to all of us.

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